Sahaj Bhadja

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Summary

Software engineer with **M.S. Computer Science** and hands-on **C/C++**, **Java**, and **Python** development across **Linux/Windows**. Experienced in **image processing**, embedded-style device software, real-time communication, and cross-platform systems with strong analytical and communication skills.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Operating Systems, Analysis of Algorithms, Computer Networks, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Operating Systems, Data Structures, Object-Oriented Programming, Computer Networks

Experience

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **image-processing** and **vision-model** workflows for handwritten and image-based submissions using OCR and multimodal models, improving grading consistency to **90%+** while handling **10K+ daily requests** under concurrent load.
- Implemented **multithreaded** batch-processing pipelines with fault-tolerant execution on **Linux**-hosted services, collaborating globally with engineers on debugging, documentation, and performance improvements across production software releases.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Developed production software services with **Python** and relational databases for 500+ users, using **Git** and **CI/CD** release workflows to deliver reliable platform updates with structured documentation and cross-team communication.
- Optimized concurrent background job execution and failure isolation, improving throughput by **3x** while maintaining **99%+ reliability** across **5K+ monthly** jobs through analytical debugging and process-level performance tuning.

MedDevice Test Systems **India**
Embedded Software Engineer Intern *January 2023 – April 2023*
- Wrote **embedded-style device software** in **Python/C** for communication-level control scenarios including pairing, reconnect, and drop/recover flows with deterministic logs, simulators, and protocol-style validation across connected hardware environments.
- Integrated automated test suites into CI pipelines on **Linux/Windows** systems with standardized coverage reports, supporting firmware-adjacent debugging and reliability improvements for regulated manufacturing device workflows.

Aasma Technology Solutions **India**
Software Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform software in **Java** and **Python** across **Windows, Linux, and macOS**, improving API response by **30%** through algorithmic optimization, caching, and low-level payload reduction for distributed system integrations.
- Maintained **20+ automated test suites** using **Git**, Jenkins, and PyTest across global build farms, collaborating with multi-discipline teams while exercising sound judgment on design changes, upgrades, and production software delivery.

Aasma Technology Solutions **India**
Software Engineer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed real-time control and synchronization software using **WebSockets**, REST APIs, and **TCP/IP** networking across **2M+ records**, delivering sub-second updates with **33% fewer conflicts** for multi-location operational control dashboards.
- Automated Docker/Jenkins deployment pipelines cutting release cycles by **50%**, creating technical documentation and analyzing system bottlenecks while supporting customer-facing software used in manufacturing and services operations.

Skills

Languages & OS: **C**, **C++**, **Java**, C# (familiar), **Python**, Linux, Windows, Unix, Bash, multithreading, concurrency, OOP, data structures, algorithms
Embedded & Imaging: embedded software, firmware development (familiar), **image processing**, computer vision, SAM, CLIP, PyTorch, real-time systems, RTOS (familiar), Embedded Linux (familiar), TCP/IP, Ethernet, serial communication (familiar), mechanical control (familiar)
Tools: **Git**, Azure DevOps (familiar), Jenkins, CI/CD, Docker, debugging, cross-platform development, AI coding assistants, technical documentation, problem solving, global team collaboration, SEMI standards (familiar)

Projects

PiXMod Live | *Image Processing, SAM, CLIP, Stable Diffusion*
- Built an **image-processing** platform with segmentation, inpainting, restoration, and content-based search using SAM, CLIP, and Stable Diffusion across **100K+** indexed embeddings, designing scalable ingestion and retrieval pipelines for vision workloads.

CaptionCraft Live | *Computer Vision, PyTorch, Flask*
- Benchmarked transformer vision architectures (BLIP, GIT, ViT-GPT-2) on **10K images / 100K captions**, improving caption accuracy from **81% to 96%** and building inference APIs for scalable real-time image-processing applications.

CPU Scheduling Simulator Live | *C++, Real-Time Scheduling, Linux Concepts*
- Implemented preemptive scheduling algorithms in **C++** with interactive controls and metric visualization across **1K+** process combinations, applying operating-system concepts relevant to real-time control and performance-critical embedded software design.